Milestone 2

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1 Project Goal

This project explores the 2024 U.S. Coast Guard / NOAA AIS dataset through an interactive visualization of major U.S. ports. Rather than treating AIS data as individual vessel trajectories, we interpret ports as dynamic mobility systems with characteristic spatial patterns, temporal rhythms, and behavioral features such as slowing down or anchoring.

AIS data provides detailed records of vessel position, speed, and identity over time, but its scale and density make direct interpretation challenging. Simple trajectory visualization often leads to visual clutter and hides higher-level structural patterns. Therefore, the goal of this project is to transform raw AIS records into an interpretable visual representation that supports exploratory analysis of maritime mobility.

In particular, we aim to study how vessel movement is organized around ports, how activity changes over time, and where congestion or stationary behavior occurs. By combining spatial, temporal, and behavioral views, the visualization highlights differences between ports and makes large-scale AIS data more interpretable. This also enables users to move beyond individual ship tracking and instead reason about aggregated traffic structures.

2 Visualization Design

Our system is designed as a multi-view interactive web visualization, combining spatial, temporal, and behavioral perspectives to analyze vessel traffic around major U.S. ports. The design follows the principle that complex spatio-temporal data should be decomposed into multiple complementary views, each emphasizing a different analytical dimension while remaining linked through interaction.

2.1 Spatial View – Traffic Density and Structure

The main interface is a map-based visualization that displays vessel activity using geographic coordinates. To handle the large scale of AIS data, density-based aggregation techniques such as hexbin or heatmap representations are used to reduce visual clutter while preserving global structure.

High-density regions highlight major ports and dominant shipping corridors, which reflect the underlying structure of maritime transportation networks. In addition, clustering-based methods are applied to identify potential port areas and anchoring zones, allowing users to distinguish between transit routes and stationary accumulation areas. This spatial abstraction helps reveal how maritime traffic is organized geographically and how different regions exhibit distinct structural patterns.

Overall, this view provides an immediate global understanding of where maritime activity is concentrated and how it varies across coastal regions.

2.2 Temporal View – Activity Over Time

The temporal view complements the spatial representation by illustrating how vessel activity evolves over time. AIS records are aggregated into hourly or daily intervals to reveal recurring patterns, fluctuations, and periodic structures in traffic intensity.

This enables the identification of peak activity periods for individual ports, as well as differences in temporal rhythms across regions. For example, some ports may exhibit strong daily cycles, while others show more irregular or continuous activity patterns. By comparing these temporal signatures,

users can better understand operational characteristics and usage intensity across different maritime hubs.

This time-oriented perspective is essential for understanding not only where maritime activity occurs, but also when it happens and how it changes dynamically.

2.3 Behavioral View – Anchoring and Waiting

To capture vessel behavior beyond spatial movement, we incorporate speed-based analysis using the Speed Over Ground (SOG) attribute. Low-speed or near-zero speed observations are interpreted as anchoring or waiting behavior, which often indicates congestion, port entry delays, or operational pauses.

These behavioral patterns are visualized as spatial clusters near ports, revealing regions where vessels accumulate or remain stationary. This provides additional insight that cannot be derived from position data alone, as it distinguishes between active movement and inactive or delayed states.

By introducing this behavioral dimension, the visualization enables a more complete understanding of maritime dynamics, particularly in terms of efficiency and congestion patterns in port areas.

2.4 Interaction Design

The visualization supports several interactive features that enable coordinated exploration across views. A time slider allows users to filter specific temporal windows, while zooming and panning support detailed spatial exploration. Additional filtering by vessel type further refines the analysis.

Importantly, interactions are linked across views, so that changes in one view (e.g., time selection or spatial zoom) are reflected in the others. This coordinated design allows users to maintain context while switching perspectives between spatial, temporal, and behavioral information.

Such an interaction model is crucial for exploring complex datasets like AIS, where insights often emerge from connecting multiple analytical dimensions rather than examining them independently.

2.5 Visualization Examples

These visualizations illustrate how different perspectives of the data reveal complementary insights.

The global density maps highlight large-scale traffic structure, showing major shipping corridors and coastal concentration of vessel movement. In contrast, the port-level views provide a more localized perspective, where fine-grained structures such as entry routes and anchoring zones become visible.

Temporal aggregation further reveals how traffic intensity changes over time. For instance, the comparison between all-hour and specific-time views (e.g., 9:00 UTC) demonstrates how activity fluctuates within a daily cycle.

Finally, vessel-type distributions provide an additional layer of analysis by characterizing the composition of traffic. Differences between global and port-level distributions highlight how certain vessel categories dominate specific regions, reflecting their operational roles.

Due to space limitations, the full interactive interface can be explored on the project website (see Section 5). Here, we present a selection of representative views illustrating the main components of the visualization.

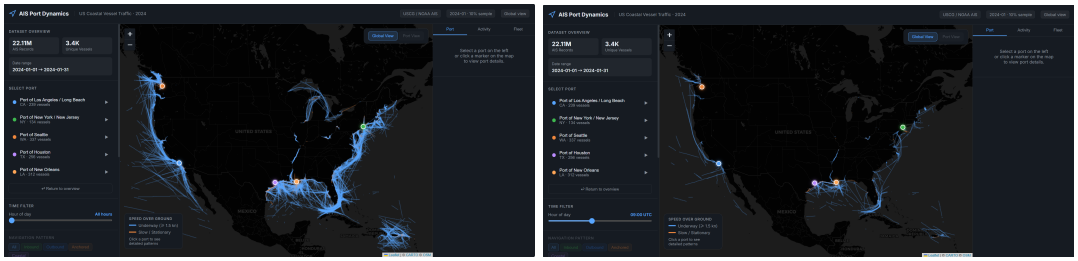

Figure 1: Global traffic density maps. Left: all hours aggregated view. Right: traffic distribution at 9:00 UTC.

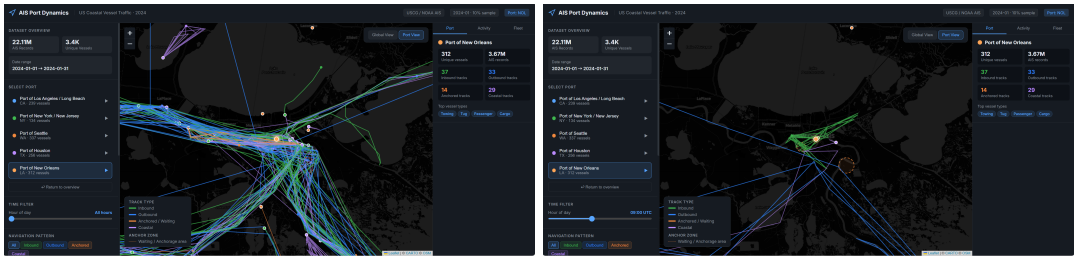

Figure 2: Port-level traffic density maps. Left: all hours aggregated view. Right: traffic distribution at 9:00 UTC.

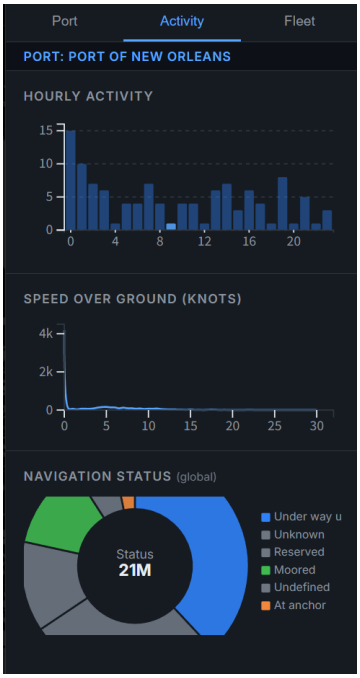

Figure 3: Temporal activity patterns showing aggregated vessel activity over time at port level.

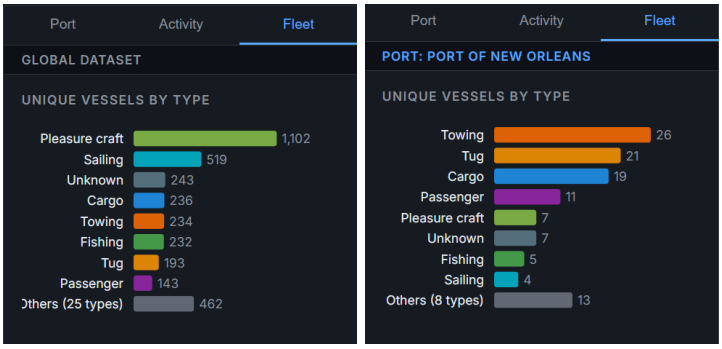

Figure 4: Distribution of unique vessels by type at global (left) and port (right) scales. Vessel categories include towing, tug, cargo, passenger, fishing, sailing, and others.

3 Extended Components

The current implementation represents an initial working prototype focused on exploratory visualization of AIS vessel activity. While it already enables basic spatial and temporal exploration, it remains

limited in terms of analytical depth and cross-view interaction. Therefore, the final system is designed to extend this prototype into a richer analytical and storytelling-oriented platform. We also aim to incorporate more specific analytical perspectives, such as port-specific movement patterns and regional differences in congestion, to reveal insights that go beyond general traffic visualization.

In the final version, we plan to introduce a more tightly integrated multi-view framework, where spatial, temporal, and behavioral views are connected through brushing and selection. This will allow users to select a subset of vessels or a specific region on the map and immediately observe how their temporal patterns and behavioral characteristics change accordingly.

Second, we aim to add a comparative analysis mode between different ports. Instead of analyzing a single region in isolation, users will be able to compare traffic structure, temporal rhythms, and congestion patterns across multiple major ports. This comparative perspective is essential for understanding regional differences in maritime operations.

Third, the system will include more advanced interaction and animation features, such as trajectory replay and time-lapse visualization. These features will enable users to observe how vessel movements evolve continuously over time, rather than only through aggregated snapshots.

Finally, we consider introducing a storytelling mode inspired by data journalism approaches. In this mode, users can follow pre-defined analytical narratives, such as comparing congestion patterns between major ports or exploring daily traffic cycles. This transforms the system from a purely exploratory tool into a guided analytical experience.

4 Related Tools and Lectures

This project builds on a combination of data processing, visualization, and web development tools. We use Python (pandas, numpy) for data cleaning and preprocessing, and Matplotlib for initial exploratory analysis. For the interactive visualization, we rely on JavaScript-based libraries such as D3.js and Plotly, along with Leaflet or Mapbox for geospatial rendering. The final interface is implemented using standard HTML and CSS.

The design and implementation are strongly informed by core topics covered in the course. Week 1–4 lectures provide the technical foundation for web-based visualization development, including JavaScript and D3.js. Week 5 on interaction, filtering, and aggregation directly informs our use of linked views and time-based filtering mechanisms. Week 6 on perception and visual encoding influences our design of density-based maps and behavioral encodings.

In addition, Week 7 on visualization design guides the overall multi-view structure of the system, while Week 8 on maps is directly applied in the spatial representation of AIS data. Finally, concepts from later lectures on storytelling with data motivate the planned evolution of the system toward narrative-driven exploration.

5 Initial Web Prototype

The initial web prototype is available at

https://com-480-data-visualization.github.io/DataVision2/

providing a first working implementation of the interactive AIS visualization system.

The current version includes the core spatial density visualization, basic temporal filtering, and initial comparative views between global and port-level activity. It serves as a proof of concept for the design principles described above and demonstrates the feasibility of handling large-scale AIS data in an interactive setting.

In addition to visualization components, the prototype also incorporates basic statistical summaries of the dataset. These include distributions of vessel types, counts of unique vessels, and aggregated activity levels over time. Such summaries provide complementary insights that support the visual exploration, helping users better interpret observed patterns and validate trends identified in the visualizations